

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location TX facility TX_way_744177375 -- station KDWH, America/Chicago
Building OSM 744177375
OSM way 744177375
Facade gap not applicable -- one building, no second facade
Weather record 42,416 real hourly records from KDWH
Plume physics NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2021-01-02 (clear_cool)
Plant limit 18.0 C
Notice required 6 h
Forecast skill 0.00 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 12 of 24 hours with 2 mode changes. 10 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 12 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent 12 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
10 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	5.550	18.0	6.110	4.123
01	FREE	yes	-	5.560	18.0	5.560	3.156
02	FREE	yes	-	6.110	18.0	5.000	3.008
03	FREE	yes	-	6.110	18.0	4.440	2.588
04	FREE	yes	-	6.660	18.0	5.000	2.735
05	FREE	yes	-	6.660	18.0	4.440	2.426
06	FREE	yes	-	6.670	18.0	5.000	2.060
07	FREE	yes	-	7.780	18.0	5.000	2.358
08	FREE	yes	-	10.000	18.0	6.110	2.918

09	FREE	yes	-	12.780	18.0	7.780	4.218
10	FREE	yes	-	16.120	18.0	8.890	5.253
11	FREE	yes	-	17.770	18.0	10.560	6.073
12	mechanical	no	dry-bulb	19.440	18.0	12.220	6.222
13	mechanical	no	dry-bulb	18.330	18.0	13.330	5.583
14	mechanical	yes	-	16.660	18.0	13.890	4.354
15	mechanical	yes	switch budget	15.560	18.0	13.890	3.453
16	mechanical	yes	switch budget	14.440	18.0	13.330	3.175
17	mechanical	yes	switch budget	13.890	18.0	9.440	3.152
18	mechanical	yes	switch budget	13.330	18.0	6.110	3.130
19	mechanical	yes	switch budget	12.770	18.0	5.560	3.347
20	mechanical	yes	switch budget	12.230	18.0	5.000	3.620
21	mechanical	yes	switch budget	12.220	18.0	4.440	4.200
22	mechanical	yes	switch budget	11.660	18.0	3.890	4.375
23	mechanical	yes	switch budget	8.330	18.0	3.890	4.429

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.550 C, which is 12.450 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.123 C, and the margin is measured, not chosen: 4.123 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.560 C, which is 12.440 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.156 C, and the margin is measured, not chosen: 3.156 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.110 C, which is 11.890 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.008 C, and the margin is measured, not chosen: 3.008 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.110 C, which is 11.890 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.588 C, and the margin is measured, not chosen: 2.588 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.660 C, which is 11.340 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.735 C, and the margin is measured, not chosen: 2.735 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.660 C, which is 11.340 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.426 C, and the margin is measured, not chosen: 2.426 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.670 C, which is 11.330 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.060 C, and the margin is measured, not chosen: 2.060 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.780 C, which is 10.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.358 C, and the margin is measured, not chosen: 2.358 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.000 C, which is 8.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.918 C, and the margin is measured, not chosen: 2.918 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.780 C, which is 5.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.218 C, and the margin is measured, not chosen: 4.218 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.120 C, which is 1.880 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.253 C, and the margin is measured, not chosen: 5.253 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

11:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.770 C, which is 0.230 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 6.073 C, and the margin is measured, not chosen: 6.073 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.440 C against a 18.0 C limit, so it fails by 1.440 C. It would take a limit 1.440 C higher, or a bound 1.440 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.330 C against a 18.0 C limit, so it fails by 0.330 C. It would take a limit 0.330 C higher, or a bound 0.330 C tighter, to change this hour.

14:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.660 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

15:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.560 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here

would cost a better one later.

16:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.440 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.890 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.330 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.770 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.230 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.220 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.660 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.330 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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